Motors and generators

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Distinguish electrical-to-mechanical transfer from mechanical-to-electrical transfer.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 12.*
A stronger field at A in that drawing convention; field lines are a model, not physical strands.

02. Worked example: explain the model
Motor electrical to mechanical; generator mechanical to electrical; both may have thermal losses.

03. Read and interpret the model
Electrical to mechanical for motor; mechanical to electrical for generator.

04. Apply the physics
60%; 40 J other transfers.

05. Explain your reasoning
Each conversion transfers some energy to surroundings; usable energy in the loop decreases.

06. Review: connect an earlier idea *Review: Unit 7, Day 3.*
Positive charge: left. Negative charge: right. Each force has magnitude 24 N.

07. Exit check
Generator.

08. STRETCH: test the model
Higher useful mechanical power at fixed input means greater efficiency; measure useful work/time and electrical V, I.

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01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 12.*
Each piece is normally a smaller magnet with both poles; cutting does not isolate a magnetic monopole.

02. Worked example: explain the model
Motor electrical to mechanical; generator mechanical to electrical; both may have thermal losses.

03. Read and interpret the model
Electrical to mechanical for motor; mechanical to electrical for generator.

04. Apply the physics
36 W; 72%; 14 W.

05. Explain your reasoning
Each conversion transfers some energy to surroundings; usable energy in the loop decreases.

06. Review: connect an earlier idea *Review: Unit 7, Day 3.*
Field becomes 1/16; assumes an isolated point charge or an exterior spherically symmetric source.

07. Exit check
$0.80 \times 0.75 = 0.60$ or 60%.

08. STRETCH: test the model
Useful 18 W; $v = P/(mg) = 0.6$ m/s; other power 6 W.